

```
7290      "IAH"    "IAD"
7290      "IAH"    "IAD"
```

Row-wise operations

You can use `map()` to apply some operations to every row. In the following example, the distance is divided by airtime and it's multiplied by 60.

```
speed = map(i -> i.Distance / i.AirTime
* 60, flights)
```

Adding a new column

A new column can be added to the existing data set with the help of the `transform` function:

```
flights = transform(flights, :Speed =>
speed)
```

This adds 'Speed' as the 22nd column to the existing flight's data set.

Let us suppose that for each destination, you want to find the count of the total flights and the number of distinct planes. This can be achieved with a `groupby` as shown below:

```
groupby((flight_count = length, plane_
count = length union), flights, :Dest,
select = :TailNum)
```

The output is shown in Figure 4.

Visualisation

One of the greatest advantages of Julia is the availability of professional grade visualisation packages, an example of which is shown below:

```
using StatsPlots
gr(fmt = :png) # choose the fast GR
backend
@df flights scatter(:DepDelay,
:ArrDelay, group = :Distance .> 1000,
layout = 2, legend = :topleft)
```

OnlineStats

OnlineStats is a popular package to compute statistics, one observation at a time. It is based on the parallel

```
julia> groupby((flight_count = length, plane_count = length union),
flights, :Dest, select = :TailNum)
Table with 116 rows, 3 columns:
Dest      flight_count  plane_count
-----
"ABQ"      2812          716
"AEX"      724           215
"AGS"       1            1
"AMA"     1297          158
"ANC"      125           38
"ASE"      125           60
"ATL"     7886          983
"AUS"     5022         1015
"AVL"      350           142
"BFL"      504           70
"BHM"     2736          616
"BKG"      110            63
"BNA"     3481          736
⋮
"SJC"      885          173
"STL"     2509          788
"TPA"     3085          697
"TUL"     2924          771
"TUS"     1565          226
"TYS"     1210          227
"VPS"      880          224
"XNA"     1172          177
```

Figure 4: *Groupby* output

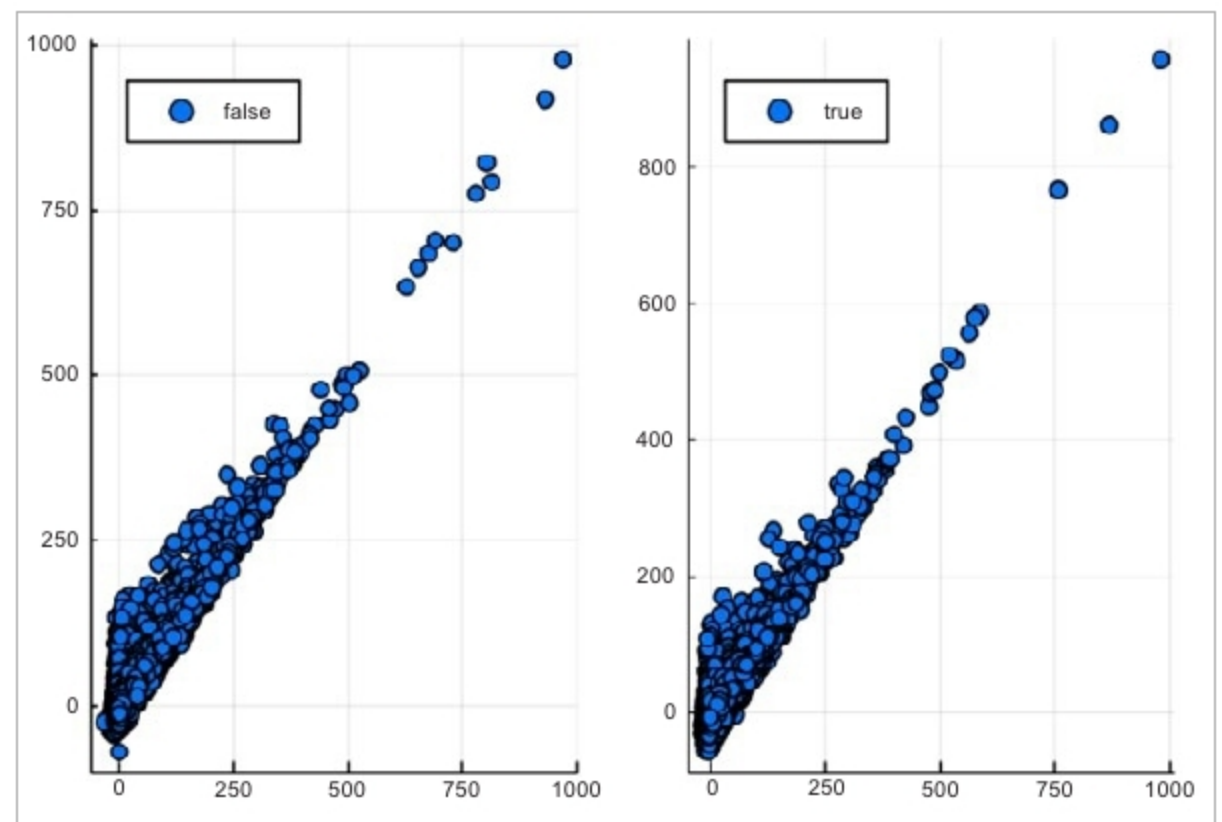


Figure 5: Visualisation

algorithms. OnlineStats can be integrated with JuliaDB in a seamless manner. Its biggest advantage is that

it can handle streaming and Big Data that does not fit in memory. Figure 6 illustrates the computation of statistics in parallel, in JuliaDB.